Kalman filtering: optimality and convergence

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1 Problem setting

Consider the discrete-time linear system

$$x_{k+1} = Ax_k + w_k, \tag{1}$$
$$y_k = Hx_k + v_k, \tag{2}$$

where

- $x_k \in \mathbb{R}^n$ is the true state, which we cannot observe directly;
- $y_k \in \mathbb{R}^m$ is the measurement available at time k;
- $A \in \mathbb{R}^{n\times n}$ is the state transition matrix, which describes how the state evolves;
- $H \in \mathbb{R}^{m\times n}$ describes how the state produces a measurement;
- w_k is the process noise;
- v_k is the measurement noise.

Intuitively, the first equation describes how the hidden state evolves over time, while the second describes what we can observe about that state through noisy measurements.

The matrices A and H are assumed to be known exactly, and the estimator uses these same matrices. The true state and the individual noise values are unknown, but we make some assumptions, those are

i) the initial state x_0 has known mean m_0 and covariance P_0, with finite second moment, that is $\mathbb{E}\left[\|x_0\|_2^2\right] < +\infty$;

ii) and for every k, the noise models' covariances, $Q \in \mathbb{R}^{n\times n}$, and $R \in \mathbb{R}^{m\times m}$, are positive definite and stationary over time, and prescribed as

$$\begin{cases} \mathbb{E}[w_k] & = 0, \\ \mathbb{E}[v_k] & = 0, \\ \mathbb{E}[w_k w_k^\top] & = Q, \\ \mathbb{E}[v_k v_k^\top] & = R. \end{cases} \tag{3}$$

iii) The initial state x_0, the process noises, and the measurement noises are mutually independent, including across time.

The objective is to estimate x_k, given by $\hat{x}_k$. Define

$$e_k := x_k - \hat{x}_k, \quad P_k := \mathbb{E}[f(e_k)], \tag{4}$$

where $P_k \in \mathbb{R}^{n\times n}$, e_k denotes the estimation error, $f : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ takes values in the positive semidefinite matrices[**?**]. Estimating $\hat{x}_k$ means minimizing P_k. Here, we will choose

$$f(e) = ee^\top.$$

This particular choice of f leads to a recursive estimator that minimizes the mean squared error ($\mathbb{E}\left[\|e_k\|_2^2\right] = \mathrm{tr}(P_k) = \mathrm{tr}(\mathbb{E}[e_k e_k^\top])$) among all affine estimators based on the entire measurement history. Under Gaussian assumptions, it is also optimal among all estimators based on that history.

Let $\hat{x}_k^-$ denote the prior estimate, i.e., the estimation of state x_k with knowledge uses from observations $y_1, \ldots, y_{k-1}$, while the $\hat{x}_k$ is gained with the additional of information from the newest observation y_k. Consequently, define

$$e_k^- := x_k - \hat{x}_k^-, \quad P_k^- := \mathbb{E}[f(e_k^-)], \tag{5}$$

where e_k^- be the error of the prior estimate, and P_k^- be the prior error covariance. Furthermore, assume that

$$\hat{x}_k^- = A\hat{x}_{k-1}, \tag{6}$$

i.e., the prior estimate can be viewed as a transition base on the previous estimated state.

We first consider updates of the form

$$\hat{x}_k = \hat{x}_k^- + K_k r_k, \tag{7}$$
$$r_k = y_k - H\hat{x}_k^-, \tag{8}$$

where $K_k \in \mathbb{R}^{n\times m}$ is a deterministic matrix to be chosen. The term r_k compares the actual measurement with the measurement predicted by the prior estimate. It is called the measurement residual. We will show that the estimator is unbiased, meaning $\mathbb{E}[e_k] = 0$, under some assumptions.

2 Error convariance

Proposition 2.1. *Assume for any given deterministic gains $K_k \in \mathbb{R}^{n\times m}$, following the updates $\hat{x}_k^- = A\hat{x}_{k-1}$ and $\hat{x}_k = \hat{x}_k^- + K_k(y_k - H\hat{x}_k^-)$. For every $k \geq 1$,*

$$e_k^- = Ae_{k-1} + w_{k-1}, \quad e_k = (I_n - K_k H)e_k^- - K_k v_k. \tag{9}$$

If $\mathbb{E}[e_0] = 0$, then $\mathbb{E}[e_k^-] = \mathbb{E}[e_k] = 0$ for every k. Consequently, P_k define in (4) is the error covariance matrix.

Proof. For $k \geq 1$, by (5) and (6), we have

$$\begin{aligned} e_k^- &= x_k - \hat{x}_k^- \\ &= x_k - A\hat{x}_{k-1}. \end{aligned}$$

Then, by (1),

$$\begin{aligned} x_k - A\hat{x}_{k-1} &= A(x_{k-1} - \hat{x}_{k-1}) + w_{k-1} \\ &= Ae_{k-1} + w_{k-1}. \end{aligned}$$

Moreover, from (2),

$$\begin{aligned} y_k - H\hat{x}_k^- &= H(x_k - \hat{x}_k^-) + v_k \\ &= He_k^- + v_k. \end{aligned}$$

Hence, by (4) and (7),

$$\begin{aligned} e_k &= x_k - \hat{x}_k \\ &= x_k - \hat{x}_k^- - K_k(y_k - H\hat{x}_k^-) \\ &= -e_k^- - K_k(He_k^- + v_k) \\ &= (I_n - K_kH)e_k^- - K_kv_k. \end{aligned}$$

Now, assume $\mathbb{E}[e_0] = 0$, we want to prove that $\mathbb{E}[e_k^-] = \mathbb{E}[e_k] = 0$. By (3), we have that

$$\begin{aligned} \mathbb{E}[e_k^-] &= A\mathbb{E}[e_{k-1}], \\ \mathbb{E}[e_k] &= (I_n - K_kH)\mathbb{E}[e_k^-]. \end{aligned}$$

The conclusion follows by induction. □

Lemma 2.2. *The error e_{k-1} is independent of w_{k-1}, and the prior error e_k^- is independent of v_k.*

Proof. Both x_{k-1} and $\hat{x}_{k-1}$ are functions of $x_0, w_0, \ldots, w_{k-2}, v_1, \ldots, v_{k-1}$ and deterministic quantities. Thus e_{k-1} is a function of variables independent of w_{k-1}. Similarly, x_k and $\hat{x}_k^-$ use only $x_0, w_0, \ldots, w_{k-1}, v_1, \ldots, v_{k-1}$, all independent of v_k. Therefore, e_{k-1} is independent of w_{k-1}, and $e_k^- = x_k - \hat{x}_k^-$ is independent of v_k, because each error is determined entirely by variables independent of the corresponding noise term. □

Proposition 2.3. *With unbiased initialization, i.e., $\mathbb{E}[\hat{x}_0] = \mathbb{E}[x_0]$, or equivalently $\mathbb{E}[e_0] = 0$, the actual covariances satisfy*

$$P_k^- = AP_{k-1}A^\top + Q, \quad P_k = (I_n - K_kH)P_k^-(I_n - K_kH)^\top + K_kRK_k^\top. \tag{10}$$

Proof. By Proposition 2.1 and (3) we have

$$\begin{aligned} P_k^- = \mathbb{E}[e_k^-(e_k^-)^\top] &= \mathbb{E}[(Ae_{k-1} + w_{k-1})(Ae_{k-1} + w_{k-1})^\top] \\ &= \mathbb{E}[Ae_{k-1}e_{k-1}^\top A^\top] + \mathbb{E}[Ae_{k-1}w_{k-1}^\top] + \mathbb{E}[w_{k-1}e_{k-1}^\top A^\top] + \mathbb{E}[w_{k-1}w_{k-1}^\top] \\ &= AP_{k-1}A^\top + A\mathbb{E}[e_{k-1}w_{k-1}^\top] + \mathbb{E}[w_{k-1}e_{k-1}^\top]A^\top + Q. \end{aligned}$$

By Lemma 2.2, we can conclude

$$P_k^- = AP_{k-1}A^\top + Q.$$

Similarly, by Proposition 2.1, (3) and let $B = (I_n - K_kH)$ we have

$$\begin{aligned} P_k = \mathbb{E}[e_ke_k^\top] &= \mathbb{E}[(Be_k^- - K_kv_k)(Be_k^- - K_kv_k)^\top] \\ &= B\mathbb{E}[e_k^-(e_k^-)^\top]B^\top + B\mathbb{E}[e_k^-v_k^\top]K_k^\top + K_k\mathbb{E}[v_k(e_k^-)^\top]B^\top + K_k\mathbb{E}[v_kv_k^\top]K_k^\top \\ &= (I_n - K_kH)P_k^-(I_n - K_kH)^\top + K_kRK_k^\top. \end{aligned}$$

□

Having expressed P_k in terms of K_k, we now seek the gain that minimizes the mean squared estimation error $\mathbb{E}\left[\|e_k\|_2^2\right] = \mathrm{tr}(P_k)$. Fix time k and a prior error covariance $C = P_k^- \succeq 0$. Choose a deterministic gain K to minimize the mean-square error of the update $\hat{x}_k(K) = \hat{x}_k^- + Kr_k$. Here, $\hat{x}_k$ is expressed as a function w.r.t K.

3 Optimality at fixed time step

Define

$$S := HCH^\top + R \succ 0, \quad K^* := CH^\top S^{-1}. \tag{11}$$

Proposition 3.1. *For every $K \in \mathbb{R}^{n\times m}$, the updated covariance satisfies*

$$P(K) = P(K^*) + (K - K^*)S(K - K^*)^\top, \tag{12}$$

where

$$P(K^*) = C - CH^\top (HCH^\top + R)^{-1} HC =: \mathcal{U}(C). \tag{13}$$

Thus, K^* is the unique minimizer of $J(K) = \mathbb{E}\|x_k - \hat{x}_k(K)\|_2^2 = \mathrm{tr}(P(K))$ and minimizes the covariance in the positive semidefinite order.

Proof. Expanding P_k, as a function P w.r.t K from (10) gives

$$\begin{aligned} P(K) &= (I_n - KH)C(I_n - KH)^\top + KRK^\top \\ &= C - KHC - CH^\top K^\top + KHCH^\top K^\top + KRK^\top \\ &= C - KHC - CH^\top K^\top + KSK^\top. \end{aligned}$$

By (11), $K^*S = CH^\top$, and $S(K^*)^\top = HC$, then

$$\begin{aligned} P(K) &= C - KS(K^*)^\top - K^*SK^\top + KSK^\top \\ &= C - K^*S(K^*)^\top + K^*S(K^*) - KS(K^*)^\top - K^*SK^\top + KSK^\top \\ &= C - CH^\top S^{-1}HC + (K - K^*)S(K - K^*)^\top \\ &= P(K^*) + (K - K^*)S(K - K^*)^\top, \end{aligned}$$

where putting K^* from (11) in (10) gives

$$\begin{aligned} P(K^*) &= (I_n - K^*H)C(I_n - K^*H)^\top + K^*R(K^*)^\top \\ &= C - K^*HC - CH^\top (K^*)^\top + K^*(HCH^\top + R)(K^*)^\top \\ &= C - K^*HC \\ &= C - CH^\top (HCH^\top + R)^{-1}HC. \end{aligned}$$

If $D = K - K^*$, then for every $u \in \mathbb{R}^n$,

$$u^\top DSD^\top u = \|S^{1/2}D^\top u\|_2^2 \geq 0,$$

so $P(K) \succeq P(K^*)$, for all $K \in \mathbb{R}^{n\times m}$. Taking traces yields

$$J(K) = J(K^*) + \|DS^{1/2}\|_F^2.$$

Because $S^{1/2}$ is invertible, equality holds exactly when $D = 0$. This proves uniqueness and global optimality. □

Corollary 3.2. *Let $J : \mathbb{R}^{n\times m} \to \mathbb{R}$,*

$$J(K) = \mathbb{E}\|x_k - \hat{x}_k(K)\|_2^2 = \mathrm{tr}(P(K)). \tag{14}$$

The function J is strictly convex in K. At the optimal gain,

$$0 \preceq P_k = \mathcal{U}(P_k^-) = P_k^- - K_k S_k K_k^\top \preceq P_k^-.$$

Moreover,

$$P_k = (I_n - K_k H)P_k^-. \tag{15}$$

Proof. For any nonzero direction D, the second derivative of $J(K+tD)$ with respect to t, with P as a function in (12),

$$\frac{\mathrm{d}}{\mathrm{d}t}J(K+tD) = 2\,\mathrm{tr}\left(\left((K+tD)S - P^{-}H^{\top}\right)D^{\top}\right),$$
$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}J(K+tD) = 2\,\mathrm{tr}(DSD^{\top}) > 0.$$

Thus, J is strictly convex in K.

At the optimal gain $K_k = K^* = P_k^{-}H^{\top}S_k^{-1}$, where $S_k = HP_k^{-}H^{\top} + R$, by (13),

$$\begin{aligned} P_k = \mathcal{U}(P_k^{-}) &= P_k^{-} - P_k^{-}H^{\top}S_k^{-1}HP_k^{-} \\ &= P_k^{-} - K_kS_kK_k^{\top}. \end{aligned}$$

The subtracted matrix $K_kS_kK_k^{\top}$ is semipositive definite, so

$$P_k \preceq P_k^{-}.$$

Furthermore, P_k can be expressed as

$$P_k = (I_n - K_kH)P_k^{-}(I_n - K_kH)^{\top} + K_kRK_k^{\top} \succeq 0.$$

Thus,

$$0 \preceq P_k \preceq P_k^{-}.$$

Moreover,

$$\begin{aligned} P_k &= P_k^{-} - P_k^{-}H^{\top}S_k^{-1}HP_k^{-} \\ &= (I_n - P_k^{-}H^{\top}S_k^{-1}H)P_k^{-} \\ &= (I_n - K_kH)P_k^{-}. \end{aligned}$$

□

Strict convexity concerns the gain at a fixed time, with a fixed prior covariance. No iterative optimization is needed, the unique minimizer is explicit. A measurement reduces modeled uncertainty relative to the prior (each individual entry does not necessarily decrease), but $P_k \preceq P_{k-1}$ need not hold, because prediction adds process noise. An easy example of this is to consider a scalar system with $A = H = 1$, $Q = R = 1$, and suppose that $P_{k-1} = 0.1$. It can be computed that $P_k^{-} = 1.1$ and $P_k = \dfrac{P_k^{-}R}{P_k^{-} + R} \approx 0.524$. Hence, $P_k \not\preceq P_{k-1}$.

So far, we have derived the following recursion which will be assumed throughout the remaining sections. Initialize $\hat{x}_0 = m_0$ and P_0 with the given initial covariance. For $k \geq 1$, compute

$$\hat{x}_k^{-} = A\hat{x}_{k-1}, \qquad P_k^{-} = AP_{k-1}A^{\top} + Q, \tag{16}$$
$$r_k = y_k - H\hat{x}_k^{-}, \qquad S_k = HP_k^{-}H^{\top} + R, \tag{17}$$
$$K_k = P_k^{-}H^{\top}S_k^{-1}, \qquad \hat{x}_k = \hat{x}_k^{-} + K_kr_k, \tag{18}$$
$$P_k = (I_n - K_kH)P_k^{-}(I_n - K_kH)^{\top} + K_kRK_k^{\top}. \tag{19}$$

We now show that, at each time k, this recursion produces the optimal affine estimator of x_k based on the entire measurement history $y_1, \ldots, y_k$.

4 Optimality over measurement history

Given the history $Y_k = (y_1^\top, \ldots, y_k^\top)^\top$, the objective is to estimate x_k by minimizing

$$\mathbb{E}\left[\|x_k - g(Y_k)\|_2^2\right],$$

i.e., finding a function g of the measurement history Y_k such that the estimate $\hat{x}_k = g(Y_k)$ minimizes the mean squared error. A one-step optimal gain need not, by itself, prove optimal use of old measurements. We now show that the Kalman estimate is the best affine function of Y_k under the original second-moment assumptions.

Lemma 4.1. *Let $z \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$ have finite second moments. An affine estimator $\tilde{z} = a + BY$ minimizes $\mathbb{E}\left[\|z - a - BY\|_2^2\right]$ if and only if*

$$\mathbb{E}[z - \tilde{z}] = 0, \quad \mathbb{E}[(z - \tilde{z})(Y - \mathbb{E}[Y])^\top] = 0. \tag{20}$$

The optimal random vector is unique up to almost-sure equality, even if its affine coefficients are not unique. In fact, its error covariance is no larger than that of any other affine estimator.

Proof. Let $e = z - \tilde{z}$ and assume that $\tilde{z}$ satisfies (20). Take any affine estimator $g(Y)$, and define

$$d := g(Y) - \tilde{z}.$$

Since d is affine in Y, it can be written as

$$d = c + D(Y - \mathbb{E}[Y]).$$

Then,

$$\mathbb{E}[ed^\top] = \mathbb{E}[e]c^\top + \mathbb{E}[e(Y - \mathbb{E}[Y])^\top]D^\top = 0.$$

The error between the estimator and true value can be written as

$$z - g(Y) = e - d.$$

Taking expectation of the error's norm gives

$$\begin{aligned}
\mathbb{E}\left[\|z - g(Y)\|_2^2\right] &= \mathbb{E}\left[\|e - d\|_2^2\right] \\
&= \mathbb{E}\left[\|e\|_2^2\right] - 2\mathbb{E}\left[e^\top d\right] + \mathbb{E}\left[\|d\|_2^2\right] \\
&= \mathbb{E}\left[\|e\|_2^2\right] + \mathbb{E}\left[\|d\|_2^2\right].
\end{aligned}$$

Thus, the mean squared error of any affine estimator equals the mean squared error of $\tilde{z}$ plus the expected squared distance between the two estimates. Moreover, the estimator $\tilde{z}$ is also unique, since if another estimator has the same minimum error, then

$$\mathbb{E}\left[\|d\|_2^2\right] = 0,$$

which implies $d = 0$ almost surely.

Conversely, assume that $\tilde{z}$ minimizes the mean squared error, and let $e = z - \tilde{z}$. For any $c \in \mathbb{R}^n$ and $t \in \mathbb{R}$, the estimator $\tilde{z} + tc$ is affine. By optimality,

$$\mathbb{E}\left[\|e - tc\|_2^2\right] \geq \mathbb{E}\left[\|e\|_2^2\right].$$

Thus, the function on the left has a minimum at $t = 0$. Differentiating at $t = 0$ gives

$$-2c^\top \mathbb{E}[e] = 0.$$

Since this holds for every $c \in \mathbb{R}^n$, we obtain $\mathbb{E}[e] = 0$. Similarly, for any $D \in \mathbb{R}^{n\times m}$, the estimator $\tilde{z}+tD(Y-\mathbb{E}[Y])$ is affine. By optimality,

$$\mathbb{E}\left[\|e - tD(Y - \mathbb{E}[Y])\|_2^2\right] \geq \mathbb{E}\left[\|e\|_2^2\right].$$

Differentiating at $t = 0$ gives

$$-2\mathbb{E}\left[e^\top D(Y - \mathbb{E}[Y])\right] = 0.$$

Since this holds for every $D \in \mathbb{R}^{n\times m}$, we obtain

$$\mathbb{E}\left[e(Y - \mathbb{E}[Y])^\top\right] = 0.$$

Therefore, $\tilde{z}$ satisfies (20). □

Theorem 4.2. *For every $k \geq 1$, let $\mathcal{A}_k = \{a + BY_k : a \in \mathbb{R}^n,\ B \in \mathbb{R}^{n\times mk}\}$ be the set of affine estimates based on Y_k. Assuming the initial mean $m_0 = \mathbb{E}[x_0]$ is known, the recursion (16)–(19) satisfies*

$$\hat{x}_k \in \operatorname*{arg\,min}_{z\in\mathcal{A}_k} \mathbb{E}\left[\|x_k - z\|_2^2\right],$$

and

$$\mathbb{E}[e_k] = 0, \quad \mathbb{E}[e_k(y_j - \mathbb{E}[y_j])^\top] = 0 \quad (1 \leq j \leq k). \tag{21}$$

The predictor $\hat{x}_k^-$ is likewise the best affine estimate of x_k from Y_{k-1}.

Proof. At time 0, the best constant estimate is $\hat{x}_0 = m_0$. In particular, $\mathbb{E}[e_0] = 0$. Suppose the conclusion holds for at time step $k-1$,

$$\mathbb{E}[e_{k-1}] = 0, \quad \mathbb{E}[e_{k-1}(y_j - \mathbb{E}[y_j])^\top] = 0 \quad (1 \leq j \leq k-1).$$

We must prove the corresponding statements at time k.

By Proposition 2.1, $\mathbb{E}[e_k] = \mathbb{E}[e_k^-] = 0$ and for $j = 1, ..., k-1$,

$$\begin{aligned}\mathbb{E}[e_k^-(y_j - \mathbb{E}[y_j])^\top] &= A\mathbb{E}[e_{k-1}(y_j - \mathbb{E}[y_j])^\top] + \mathbb{E}[w_{k-1}(y_j - \mathbb{E}[y_j])^\top] \\ &= 0,\end{aligned}$$

since the first term vanishes by induction assumption, the second term vanishes since the process noise and the measurement are uncorrelated. Also, $\hat{x}_k^- = A\hat{x}_{k-1}$ is affine in Y_{k-1}, since $\hat{x}_{k-1}$ is. Thus, by Lemma 4.1, $\hat{x}_k^-$ is a best affine estimate of x_k from Y_{k-1}.

By definition,

$$r_k = y_k - H\hat{x}_k^-.$$

Using $y_k = Hx_k + v_k$, we get

$$r_k = He_k^- + v_k.$$

Then, we have

$$\begin{aligned}\mathbb{E}[r_k] &= H\mathbb{E}[e_k^-] + \mathbb{E}[v_k] \\ &= 0.\end{aligned}$$

and for $j = 1, ..., k-1$,

$$\begin{aligned}\mathbb{E}[r_k(y_j - \mathbb{E}[y_j])^\top] &= H\mathbb{E}[e_k^-(y_j - \mathbb{E}[y_j])^\top] + \mathbb{E}[v_k(y_j - \mathbb{E}[y_j])^\top] \\ &= 0,\end{aligned}$$

since v_k is measurement noise of time step k, so it is unrelated to measurement from time step $1, ..., k-1$. Thus, r_k contains the part of the current measurement that is uncorrelated with the earlier measurements. This is why it is also called the innovation.

Using $\mathbb{E}[e_k^-(e_k^-)^\top] = P_k^-$ and Lemma 2.2,

$$\begin{aligned}
\mathbb{E}[r_k r_k^\top] &= \mathbb{E}[(He_k^- + v_k)(He_k^- + v_k)^\top] \\
&= H\mathbb{E}[e_k^-(e_k^-)^\top]H^\top + H\mathbb{E}[e_k^- v_k^\top] + \mathbb{E}[v_k(e_k^-)^\top]H^\top + \mathbb{E}[v_k v_k^\top] \\
&= HP_k^-H^\top + R = S_k, \\
\mathbb{E}[e_k^- r_k^\top] &= \mathbb{E}[e_k^-(He_k^- + v_k)^\top] \\
&= \mathbb{E}[e_k^-(e_k^-)^\top]H^\top + \mathbb{E}[e_k^- v_k^\top] \\
&= P_k^-H^\top.
\end{aligned}$$

The update is

$$\hat{x}_k = \hat{x}_k^- + K_k r_k,$$

so

$$e_k = e_k^- - K_k r_k.$$

Therefore,

$$\begin{aligned}
\mathbb{E}[e_k r_k^\top] &= \mathbb{E}[e_k^- r_k^\top] - K_k\mathbb{E}[r_k r_k^\top] \\
&= P_k^-H^\top - K_k S_k.
\end{aligned}$$

The choice

$$K_k = P_k^-H^\top S_k^{-1},$$

makes this expression zero.

Furthermore, for $j = 1, ..., k-1$,

$$\begin{aligned}
\mathbb{E}[e_k(y_j - \mathbb{E}[y_j])^\top] &= \mathbb{E}[e_k^-(y_j - \mathbb{E}[y_j])^\top] + K_k\mathbb{E}[r_k(y_j - \mathbb{E}[y_j])^\top] \\
&= 0.
\end{aligned}$$

We have done all the preparation part. So far, we know that e_k is uncorrelated with every earlier measurement and the current residual r_k. But the theorem requires orthogonality to the current measurement y_k as well.

From the residual definition,

$$y_k = H\hat{x}_k^- + r_k.$$

Centering gives

$$y_k - \mathbb{E}[y_k] = H(\hat{x}_k^- - \mathbb{E}[\hat{x}_k^-]) + r_k.$$

Since $\hat{x}_k^-$ is affine in earlier observations, there exist matrices $B_j \in \mathbb{R}^{n\times m}$ and a vector $a \in \mathbb{R}^n$ such that

$$\hat{x}_k^- = a + \sum_{j=1}^{k-1} B_j y_j.$$

Centering gives

$$\hat{x}_k^- - \mathbb{E}[\hat{x}_k^-] = \sum_{j=1}^{k-1} B_j(y_j - \mathbb{E}[y_j]).$$

Hence,

$$y_k - \mathbb{E}[y_k] = \sum_{j=1}^{k-1} HB_j(y_j - \mathbb{E}[y_j]) + r_k.$$

Finally,

$$\begin{aligned}\mathbb{E}[e_k(y_k - \mathbb{E}[y_k])^\top] &= \sum_{j=1}^{k-1} \mathbb{E}[e_k(y_j - \mathbb{E}[y_j])^\top] B_j^\top H^\top + \mathbb{E}[e_k r_k^\top] \\ &= 0.\end{aligned}$$

Since $\hat{x}_k$ is affine in Y_k, by Lemma 4.1, $\hat{x}_k$ is the best affine estimate of x_k from Y_{k-1}. □

In this second-moment setting, estimation residuals are uncorrelated, they need not be independent. The recursion preserves everything in the old history needed for the best affine estimate of the current state. Without Gaussianity, nonlinear functions of that history may still improve the estimate.

5 Gaussian filtering and unrestricted optimality

For this section, assume that x_0 and all process and measurement noises are Gaussian, while retaining their mutual independence. Gaussian distributions with singular covariance matrices are allowed. We want to show that, under these Gaussian assumptions, the estimate $\hat{x}_k$ defined by (18) minimizes the mean squared error among all measurable estimators based on the measurement history Y_k, not only among affine estimators.

Lemma 5.1. *Suppose (z, t) is jointly Gaussian with mean (a, b) and block covariance*

$$\begin{pmatrix} C & D \\ D^\top & S \end{pmatrix}, \quad S \succ 0.$$

Then

$$z \mid t \sim \mathcal{N}\big(a + DS^{-1}(t - b),\ C - DS^{-1}D^\top\big). \tag{22}$$

Proof. Let $\epsilon = z - a - DS^{-1}(t - b)$. Equivalently,

$$z = a + DS^{-1}(t - b) + \epsilon.$$

Then,

$$\begin{aligned}\mathbb{E}[\epsilon] &= \mathbb{E}[z] - a - DS^{-1}(\mathbb{E}[t] - b) \\ &= 0, \\ \mathrm{Cov}(\epsilon, t) &= \mathrm{Cov}(z - a - DS^{-1}(t - b), t) \\ &= \mathrm{Cov}(z, t) - \mathrm{Cov}(a, t) - DS^{-1}(\mathrm{Cov}(t, t) - \mathrm{Cov}(b, t)) \\ &= D - 0 - DS^{-1}(S - 0) \\ &= 0.\end{aligned}$$

Thus, the residual ϵ has zero mean and zero cross covariance with the observation t. Moreover,

$$\begin{aligned}\mathrm{Cov}(\epsilon) &= \mathbb{E}[(z - a - DS^{-1}(t - b))(z - a - DS^{-1}(t - b))^\top] \\ &= \mathbb{E}[(z - a)(z - a)^\top] - \mathbb{E}[(z - a)(t - b)^\top](DS^{-1})^\top \\ &\quad - DS^{-1}\mathbb{E}[(t - b)(z - a)^\top] + DS^{-1}\mathbb{E}[(t - b)(t - b)^\top](DS^{-1})^\top \\ &= C - D(S^{-1})^\top D^\top - DS^{-1}D^\top + DS^{-1}S(S^{-1})^\top D^\top \\ &= C - DS^{-1}D^\top.\end{aligned}$$

Since (z,t) is jointly Gaussian and (ϵ,t) is an affine transformation of (z,t), the pair (ϵ,t) is also jointly Gaussian. Its covariance has the block form

$$\begin{pmatrix} C - DS^{-1}D^\top & 0 \\ 0 & S \end{pmatrix},$$

and uncorrelation implies independence.

If we observe $t = t_0$, then

$$z = a + DS^{-1}(t_0 - b) + \epsilon.$$

The first term is now a fixed vector. Independence means observing t does not change the distribution of ϵ, so

$$\epsilon \mid t = t_0 \sim \mathcal{N}(0, C - DS^{-1}D^\top).$$

Thus,

$$z \mid t \sim \mathcal{N}\big(a + DS^{-1}(t-b),\ C - DS^{-1}D^\top\big).$$

□

Theorem 5.2. *Let $\mathcal{F}_k = \sigma(y_1, \dots, y_k)$ and let $\mathcal{F}_0$ be the trivial sigma-algebra. For every $k \geq 1$,*

$$x_k \mid \mathcal{F}_{k-1} \sim \mathcal{N}(\hat{x}_k^-, P_k^-), \quad x_k \mid \mathcal{F}_k \sim \mathcal{N}(\hat{x}_k, P_k). \tag{23}$$

In particular,

$$\hat{x}_k = \mathbb{E}[x_k \mid \mathcal{F}_k], \quad \mathrm{Cov}(x_k \mid \mathcal{F}_k) = P_k. \tag{24}$$

Proof. The initialization is $x_0 \sim \mathcal{N}(m_0, P_0)$. Assume at time $k-1$, we have

$$x_{k-1} \mid \mathcal{F}_{k-1} \sim \mathcal{N}(\hat{x}_{k-1}, P_{k-1}).$$

Then

$$Ax_{k-1} \mid \mathcal{F}_{k-1} \sim \mathcal{N}(A\hat{x}_{k-1}, AP_{k-1}A^\top).$$

Furthermore, w_{k-1} is independent of both x_{k-1} and the past measurements. Therefore,

$$Ax_{k-1} + w_{k-1} \mid \mathcal{F}_{k-1} \sim \mathcal{N}(A\hat{x}_{k-1}, AP_{k-1}A^\top + Q).$$

Hence,

$$x_k \mid \mathcal{F}_{k-1} \sim \mathcal{N}(\hat{x}_k^-, P_k^-).$$

The measurement equation is

$$y_k = Hx_k + v_k.$$

Conditionally on $\mathcal{F}_{k-1}$, the vectors x_k and v_k are independent Gaussian vectors. Moreover,

$$\begin{pmatrix} x_k \\ y_k \end{pmatrix} = \begin{pmatrix} I & 0 \\ H & I \end{pmatrix} \begin{pmatrix} x_k \\ v_k \end{pmatrix},$$

i.e., the pair (x_k, y_k) is therefore a linear transformation of a Gaussian vector, so it is jointly Gaussian.

From Theorem 4.2, we have for $e_k^- = x_k - \hat{x}_k^-$, $r_k = y_k - H\hat{x}_k^-$,

$$\begin{aligned} \mathbb{E}[r_k \mid \mathcal{F}_{k-1}] &= 0, \\ \mathbb{E}[r_k r_k^\top \mid \mathcal{F}_{k-1}] &= HP_k^-H + R, \\ \mathbb{E}[e_k^- r_k^\top \mid \mathcal{F}_{k-1}] &= P_k^- H^\top. \end{aligned}$$

Thus,

$$\begin{aligned}\mathbb{E}[y_k \mid \mathcal{F}_{k-1}] &= H\hat{x}_k^-,\\ \mathrm{Cov}(y_k \mid \mathcal{F}_{k-1}) &= HP_k^-H^\top + R,\\ \mathrm{Cov}(x_k, y_k \mid \mathcal{F}_{k-1}) &= P_k^-H^\top\end{aligned}$$

Conditionally on $\mathcal{F}_{k-1}$, the pair (x_k, y_k) is jointly Gaussian with mean and covariance

$$\begin{pmatrix}\hat{x}_k^- \\ H\hat{x}_k^-\end{pmatrix}, \quad \begin{pmatrix}P_k^- & P_k^-H^\top \\ HP_k^- & S_k\end{pmatrix},$$

where $S_k = HP_k^-H + R$. Then, by Lemma 5.1, conditioning on $\mathcal{F}_{k-1}$

$$x_k \mid y_k \sim \mathcal{N}(\hat{x}_k^- + P_k^-HS_k^{-1}(y_k - H\hat{x}_k^-), P_k^- - P_kH^\top S_k^{-1}HP_k^-),$$

or

$$x_k \mid \mathcal{F}_k \sim \mathcal{N}(\hat{x}_k, P_k).$$

□

Lemma 5.3. *For any square-integrable z and sigma-algebra $\mathcal{F}$, set $\bar{z} = \mathbb{E}[z \mid \mathcal{F}]$. For every square-integrable, $\mathcal{F}$-measurable random vector g,*

$$\mathbb{E}\left[\|z - g\|_2^2\right] = \mathbb{E}\left[\|z - \bar{z}\|_2^2\right] + \mathbb{E}\left[\|\bar{z} - g\|_2^2\right]. \tag{25}$$

Proof. Expand $z - g = (z - \bar{z}) + (\bar{z} - g)$. The cross term vanishes, because

$$\mathbb{E}[(z - \bar{z})^\top(\bar{z} - g)] = \mathbb{E}\big[\mathbb{E}[z - \bar{z} \mid \mathcal{F}]^\top(\bar{z} - g)\big] = 0.$$

□

Corollary 5.4. *Under Gaussianity, for every measurable $g(Y_k)$ with finite mean-square error,*

$$\mathbb{E}\left[\|x_k - g(Y_k)\|_2^2\right] = \mathrm{tr}(P_k) + \mathbb{E}\left[\|\hat{x}_k - g(Y_k)\|_2^2\right] \geq \mathrm{tr}(P_k).$$

Thus the Kalman estimate is the unique optimal random vector up to almost-sure equality, among all such estimators.

Proof. We have

$$\|g(Y_k)\|_2^2 = \|x_k - (x_k - g(Y_k))\|_2^2 \leq 2\|x_k\|_2^2 + 2\|x_k - g(Y_k)\|_2^2,$$

and both terms on the right have finite expectation. Thus, $g(Y_k)$ is square-integrable. By Lemma 5.3, with $\bar{z} = \mathbb{E}[x_k \mid \mathcal{F}_k] = \hat{x}_k$,

$$\mathbb{E}\left[\|x_k - g(Y_k)\|_2^2\right] = \mathbb{E}\left[\|x_k - \hat{x}_k\|_2^2\right] + \mathbb{E}\left[\|\hat{x}_k - g(Y_k)\|_2^2\right] \geq \mathbb{E}\left[\|x_k - \hat{x}_k\|_2^2\right] = \mathrm{tr}(P_k).$$

□

This optimality is exact at every finite time; it is not a convergence statement. Gaussianity makes the conditional mean affine, which is why the best affine estimate also solves the unrestricted problem. Lemma 5.3 itself does not require Gaussianity.

6 Convergence of the covariance and gain

We want to prove convergence assumptions of finite second moments and

$$Q \succ 0, \quad R \succ 0, \quad (A, H) \text{ observable.} \tag{26}$$

Definition 6.1 (Observability and Schur stability)**.** The pair (A, H) is observable if

$$\mathcal{O}_n = \begin{pmatrix} H \\ HA \\ \vdots \\ HA^{n-1} \end{pmatrix} \in \mathbb{R}^{mn \times n} \tag{27}$$

has rank n. A square matrix G is Schur stable if every eigenvalue has modulus less than 1, equivalently, $G^j \xrightarrow[j\to\infty]{} 0$.

To separate prior and updated covariances, write $C_k = P_k^-$ and use the update map $\mathcal{U}$ defined in (13). Then

$$C_1 = AP_0A^\top + Q, \quad C_{k+1} = \mathcal{T}(C_k) := A\mathcal{U}(C_k)A^\top + Q. \tag{28}$$

For any $C \succeq 0$, let

$$K(C) = CH^\top(HCH^\top + R)^{-1}, \quad L(C) = AK(C). \tag{29}$$

Lemma 6.2. *Let $\Psi_L : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$,*

$$\Psi_L(C) = (A - LH)C(A - LH)^\top + Q + LRL^\top. \tag{30}$$

Then

$$\Psi_L(C) = \mathcal{T}(C) + (L - L(C))(HCH^\top + R)(L - L(C))^\top. \tag{31}$$

Consequently, $\mathcal{T}(C) \preceq \Psi_L(C)$ for every L, with equality at $L = L(C)$, and $\mathcal{T}$ is continuous and order preserving.

Proof. By definition,

$$\mathcal{T}(C) = ACA^\top - ACH^\top(HCH^\top + R)^{-1}HCA^\top + Q.$$

Expanding Ψ_L,

$$\begin{aligned}
\Psi_L(C) &= ACA^\top - LHCA^\top - ACH^\top L^\top + LHCH^\top L^\top + Q + LRL^\top \\
&= ACA^\top - L(C)HCA^\top + Q + L(C)HCA^\top - LHCA^\top - ACH^\top L^\top + L(HCH^\top + R)L^\top \\
&= \mathcal{T}(C) + L(C)(HCH^\top + R)L(C)^\top - L(HCH^\top + R)L(C)^\top - L(C)(HCH^\top + R)L^\top \\
&+ L(HCH^\top + R)L^\top, \\
&= \mathcal{T}(C) + (L - L(C))(HCH^\top + R)(L - L(C))^\top.
\end{aligned}$$

because

$$\begin{aligned}
ACH^\top &= L(C)(HCH^\top + R), \\
HCA^\top &= (HCH^\top + R)L(C)^\top.
\end{aligned}$$

Since $HCH^\top + R \succ 0$, the additional term is positive semidefinite. For every $u \in \mathbb{R}^n$

$$u^\top(L - L(C))(HCH^\top + R)(L - L(C))^\top u = \left\|(HCH^\top + R)^{1/2}(L - L(C))^\top u\right\|_2^2 \geq 0.$$

Consequently,

$$\mathcal{T}(C) \preceq \Psi_L(C).$$

Choosing $L = L(C)$ makes the additional term zero, i.e., $\mathcal{T}(C) = \Psi_{L(C)}(C)$. Let $M, N \in \mathbb{R}^{n\times n}$ such that $M \preceq N$. Then, for every L,

$$\Psi_L(M) - \Psi_L(N) = (A - LH)(M - N)(A - LH)^\top \preceq 0.$$

Therefore,

$$\Psi_L(M) \preceq \Psi_L(N).$$

Choose $L = L(N)$. Then

$$\mathcal{T}(M) \preceq \Psi_{L(N)}(M), \quad \Psi_{L(N)}(M) \preceq \Psi_{L(N)}(N), \quad \Psi_{L(N)}(N) = \mathcal{T}(N).$$

Hence, $\mathcal{T}$ is order preserving.

Finally, the map $C \mapsto HCH^\top + R$ is continuous. Since $R \succ 0$, for every $C \succeq 0$ we have $HCH^\top + R \succ 0$. Matrix inversion is continuous on the set of invertible matrices, so $C \mapsto (HCH^\top + R)^{-1}$ is continuous. Therefore,

$$\mathcal{T}(C) = ACA^\top - ACH^\top(HCH^\top + R)^{-1}HCA^\top + Q$$

is continuous, as sums and products of continuous matrix-valued maps are continuous. □

Lemma 6.3. *If (A, H) is observable, there exists a finite matrix $B \succeq 0$, depending only on A, H, Q, R, such that*

$$\mathcal{T}^n(C) \preceq B \quad \textit{for every } C \succeq 0.$$

Here $\mathcal{T}^n$ means n successive applications of $\mathcal{T}$ and it is the prior covariance error as seen from (28).

Proof. Consider the auxiliary model

$$z_{i+1} = Az_i + w_i, \quad y_i = Hz_i + v_i,$$

with a random initial state z_0 satisfying

$$\mathbb{E}[z_0] = 0, \quad \mathrm{Cov}(z_0) = C,$$

independent of the future noises, where $C \succeq 0$ is arbitrary. Expanding the recursion gives

$$z_i = A^i z_0 + \sum_{\ell=0}^{i-1} A^{i-1-\ell} w_\ell.$$

Therefore,

$$y_i = HA^i z_0 + \sum_{\ell=0}^{i-1} HA^{i-1-\ell} w_\ell + v_i.$$

Let $\eta_i = \sum_{\ell=0}^{i-1} HA^{i-1-\ell} w_\ell + v_i$. Stacking the first n observations gives

$$Z = \mathcal{O}_n z_0 + \eta,$$

where

$$\mathcal{O}_n = \begin{pmatrix} H \\ HA \\ \vdots \\ HA^{n-1} \end{pmatrix}, \quad Z = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta_0 \\ \eta_1 \\ \vdots \\ \eta_{n-1} \end{pmatrix}.$$

By definition, (A, H) is observable meaning $\operatorname{rank}(\mathcal{O}_n) = n$. Consequently, $\mathcal{O}_n^\top \mathcal{O}_n$ is invertible, and

$$D = (\mathcal{O}_n^\top \mathcal{O}_n)^{-1} \mathcal{O}_n^\top \in \mathbb{R}^{n \times nm}$$

satisfies

$$D\mathcal{O}_n = I_n.$$

Applying D to the observations yields

$$DZ = D\mathcal{O}_n z_0 + D\eta = z_0 + D\eta.$$

Thus DZ is an estimate of z_0 with error

$$z_0 - DZ = -D\eta.$$

This error contains only noises, it contains no z_0 related terms. Hence its covariance does not depend on the initial covariance C.

A true state at time n is given by

$$z_n = A^n z_0 + \sum_{\ell=0}^{n-1} A^{n-1-\ell} w_\ell.$$

Choose the simple predictor

$$\tilde{z}_n = A^n DZ.$$

Because $DZ = z_0 + D\eta$,

$$\tilde{z}_n = A^n z_0 + A^n D\eta.$$

Subtracting gives

$$z_n - \tilde{z}_n = \sum_{\ell=0}^{n-1} A^{n-1-\ell} w_\ell - A^n D\eta =: \xi.$$

Therefore,

$$B := \operatorname{Cov}(\xi)$$

depends only on A, H, Q, R, and not on C. It is finite because ξ is a finite linear combination of noises with finite covariances.

The predictor $\tilde{z}_n$ is linear in Z, so it is a possible affine predictor. Let $\hat{z}_n$ be the Kalman predictor from the same observations. The affine projection property gives

$$\operatorname{Cov}(z_n - \tilde{z}_n) = \operatorname{Cov}(z_n - \hat{z}_n) + \operatorname{Cov}(\hat{z}_n - \tilde{z}_n).$$

Since the last matrix is a covariance matrix, so it is positive semidefinite,

$$\operatorname{Cov}(z_n - \hat{z}_n) \preceq \operatorname{Cov}(z_n - \tilde{z}_n).$$

Repalce $\operatorname{Cov}(z_n - \hat{z}_n)$ with $\mathcal{T}^n(C)$, and $\operatorname{Cov}(z_n - \tilde{z}_n)$ with B, we got

$$\mathcal{T}^n(C) \preceq B \quad \text{for every } C \succeq 0.$$

□

Lemma 6.4. *Under* (26)*, the sequence* $X_0 = 0$*,* $X_{j+1} = \mathcal{T}(X_j)$ *increases to a finite limit* $C_\infty \succ 0$*, and*

$$C_\infty = AC_\infty A^\top + Q - AC_\infty H^\top (HC_\infty H^\top + R)^{-1} HC_\infty A^\top. \tag{32}$$

Proof. Since $X_1 = Q \succeq X_0$, applying Lemma 6.2, we have

$$X_2 = \mathcal{T}(X_1) \succeq \mathcal{T}(X_0) = X_1.$$

Thus,

$$0 = X_0 \preceq X_1 \preceq X_2 \preceq \cdots.$$

By Lemma 6.3, there exists a finite matrix B such that

$$\mathcal{T}^n(X_0) \preceq B.$$

Consequently,

$$0 \preceq X_j \preceq B,$$

for every j. By Vigier's Theorem [1, Theorem 3.18, p.44], there exists a symmetric matrix C_∞ such that for any $u \in \mathbb{R}^n$, we have

$$X_j u \xrightarrow[j\to\infty]{} C_\infty u.$$

Set $D_j = X_j - C_\infty$. Taking the basis vectors $e_1, \ldots, e_n$ of $\mathbb{R}^n$, convergence gives

$$\|D_j e_i\|_2 \xrightarrow[j\to\infty]{} 0, \quad (i = 1, \ldots, n).$$

Consequently,

$$\|D_j\|_F^2 = \sum_{i=1}^{n} \|D_j e_i\|_2^2 \xrightarrow[j\to\infty]{} 0.$$

Thus $X_j \to C_\infty$ in Frobenius norm, and hence in every matrix norm because the matrix space is finite-dimensional. Since $X_j \succeq Q$ for $j \geq 1$, $C_\infty \succeq Q \succ 0$. Continuity of $\mathcal{T}$ gives $C_\infty = \mathcal{T}(C_\infty)$, which is (32). □

Lemma 6.5. *Let* $K_\infty = K(C_\infty)$*,* $L_\infty = AK_\infty$*, and* $G_\infty = A - L_\infty H = A(I_n - K_\infty H)$*. Then* G_∞ *is Schur stable.*

Proof. Equality in (31) at the convergence point yields

$$C_\infty = G_\infty C_\infty G_\infty^\top + D_\infty, \quad D_\infty = Q + L_\infty R L_\infty^\top \succ 0. \tag{33}$$

Take an eigenvalue λ of $G^\top$, with a nonzero eigenvector u,

$$G_\infty^\top u = \lambda u.$$

Here, λ and u can take complex value. Taking conjugate transposes gives

$$u^* G_\infty = u^* \bar{\lambda}.$$

Then, we have

$$\begin{aligned} u^* C u &= u^* G C G^\top u + u^* D u \\ &= |\lambda|^2 u^* C u + u^* D u. \end{aligned}$$

Rearranging gives

$$(1 - |\lambda|^2)u^*Cu = u^*Du.$$

Because $u \neq 0$ and $C, D \succ 0$,

$$1 - |\lambda|^2 = \frac{u^*Du}{u^*Cu} > 0,$$

which implies

$$|\lambda| < 1.$$

This holds for every eigenvalue of $G^\top$. Thus, G is Schur stable, and consequently $G^j \xrightarrow[j\to\infty]{} 0$. □

Theorem 6.6. *Under* (26)*, C_∞ is the unique positive semidefinite solution of* (32)*. For every $P_0 \succeq 0$, the filtering recursion satisfies*

$$P_k^- \longrightarrow C_\infty, \quad P_k \longrightarrow P_\infty := \mathcal{U}(C_\infty), \quad K_k \longrightarrow K_\infty. \tag{34}$$

Moreover $P_\infty \succ 0$, and the updated covariance satisfies the equivalent fixed-point equation

$$P_\infty = C_\infty - C_\infty H^\top (HC_\infty H^\top + R)^{-1} HC_\infty, \quad C_\infty = AP_\infty A^\top + Q. \tag{35}$$

Proof. In Lemma 6.4, we established that starting at $X_0 = 0$, $X_{j+1} = \mathcal{T}(X_j) \xrightarrow[j\to\infty]{} C_\infty$. Now, consider $Z_0 \succeq 0 = X_0$, then by monotonicity of $\mathcal{T}$ from Lemma 6.2,

$$X_j \preceq Z_j,$$

for every j. From Lemma 6.2, fix L at L_∞, and using (33), for any Z

$$\mathcal{T}(Z) \preceq \Psi_{L_\infty}(Z) = C_\infty + G_\infty(Z - C_\infty)G_\infty^\top.$$

Iterating this inequality gives

$$X_j \preceq Z_j \preceq C_\infty + G_\infty^j(Z_0 - C_\infty)(G_\infty^\top)^j.$$

By Lemma 6.4, the lower bound tends to C_∞, and the upper bound does so by Lemma 6.5. Hence, taking quadratic form and polarizing gives

$$Z_j \xrightarrow[j\to\infty]{} C_\infty.$$

This proves convergence for every initial prior covariance, and converge to C_∞ in particular for $C_1 = AP_0A^\top + Q$. Thus, prove uniqueness of C_∞. Continuity of $\mathcal{U}$ and $K(C)$ gives

$$P_k = \mathcal{U}(C_k) \longrightarrow \mathcal{U}(C_\infty) = P_\infty, \quad K_k = K(C_k) \longrightarrow K(C_\infty) = K_\infty.$$

Finally, for $C \succ 0$ the matrix inversion identity[2, Section 11.7, p.140] gives

$$\mathcal{U}(C) = (C^{-1} + H^\top R^{-1} H)^{-1}.$$

The matrix inside inverse is positive definite ($R \succ 0$, $C \succeq 0$), so $P_\infty = \mathcal{U}(C_\infty) \succ 0$. □

Corollary 6.7. *With the original unbiased initialization,*

$$\sup_{k\geq 0} \mathbb{E}\left[\|e_k\|^2\right] < \infty, \quad \mathbb{E}\left[\|e_k\|^2\right] \longrightarrow \operatorname{tr}(P_\infty) > 0.$$

Proof. Use $\mathbb{E}\left[\|e_k\|_2^2\right] = \operatorname{tr}(P_k)$, convergence in Theorem 6.6, and $P_\infty \succ 0$. □

Recall the main recursion we have drawn. Initialize $\hat{x}_0 = m_0$ and P_0 with the given initial covariance. For $k \geq 1$, compute

$$\hat{x}_k^- = A\hat{x}_{k-1}, \qquad P_k^- = AP_{k-1}A^\top + Q, \tag{36}$$

$$r_k = y_k - H\hat{x}_k^-, \qquad S_k = HP_k^-H^\top + R, \tag{37}$$

$$K_k = P_k^- H^\top S_k^{-1}, \qquad \hat{x}_k = \hat{x}_k^- + K_k r_k, \tag{38}$$

$$P_k = (I_n - K_k H)P_k^-(I_n - K_k H)^\top + K_k R K_k^\top. \tag{39}$$

Under the strong assumption (26), we have proved that

$$P_k^- \longrightarrow C_\infty, \quad P_k \longrightarrow P_\infty, \quad K_k \longrightarrow K_\infty.$$

However, the familiar general Riccati theorem allows $Q \succeq 0$ when (A, H) is detectable and $(A, Q^{1/2})$ is stabilizable. Detectability means that every mode not visible through H is Schur stable. Stabilizability of (A, B) means that every mode not reachable through the input matrix B is Schur stable. These can be expressed by the rank conditions

$$\operatorname{rank}\begin{pmatrix} \lambda I_n - A \\ H \end{pmatrix} = n, \quad \operatorname{rank}\begin{pmatrix} \lambda I_n - A & B \end{pmatrix} = n \quad (|\lambda| \geq 1), \tag{40}$$

respectively, over complex λ. This broader theorem is a standard extension, discussed in [3, Theorem 3.1, p.5 and Section 4, p.6]. In particular, observability is not needed for the finite-time optimality results of Sections 3–5.

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